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ON THE CONTRACTION PROPERTIES OF SINKHORN SEMIGROUPS

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Abstract. We develop a novel stability theory for Sinkhorn semigroups based on Lyapunov techniques and quantitative contraction coefficients, and establish exponential convergence of Sinkhorn iterations on weighted Banach spaces. This operator-theoretic framework yields explicit exponential decay rates of Sinkhorn iterates toward Schrödinger bridges with respect to a broad class of ϕ-divergences and Kantorovich-type distances, including relative entropy, squared Hellinger integrals, α-divergences, weighted total variation norms, and Wasserstein distances. To the best of our knowledge, these results provide the first systematic contraction inequalities of this kind for entropic transport and the Sinkhorn algorithm.

We further introduce Lyapunov contraction principles under minimal regularity assumptions, leading to quantitative exponential stability estimates for a large family of Sinkhorn semigroups. The framework applies to models with polynomially growing potentials and heavy-tailed marginals on general normed spaces, as well as to more structured boundary state-space models, including semicircle transitions and Beta, Weibull, and exponential marginals, together with semi-compact settings. Finally, our approach extends naturally to statistical finite mixtures of such models, including kernel-based density estimators arising in modern generative modeling.

1. Introduction

1.1. **Entropic optimal transport.** Let $(\mathbb{X},\lambda)$ and $(\mathbb{Y},\nu)$ be locally compact Polish spaces equipped with locally bounded positive measures λ and ν. We consider probability measures of the form

$$\lambda_U(dx) := e^{-U(x)}\ \lambda(dx) \quad \text{and} \quad \nu_V(dy) := e^{-V(y)}\ \nu(dy) \tag{1.1}$$

where $U:\mathbb{X}\to\mathbb{R}$ and $V:\mathbb{Y}\to\mathbb{R}$ are measurable potential functions chosen so that λ_U and ν_V are normalized. Let $\Pi_{U,V}$ be the set of probability measures $\mathcal{P}$ on $(\mathbb{X}\times\mathbb{Y})$, with prescribed marginals (λ_U,ν_V). We introduce a reference Markov kernel $\mathcal{Q}$ from $\mathbb{X}$ to $\mathbb{Y}$ of the form

$$\mathcal{Q}(x,dy) = e^{-W(x,y)}\ \nu(dy) \quad \text{and set} \quad \mathcal{R}(y,dx) := e^{-W(x,y)}\ \lambda(dx). \tag{1.2}$$

where $W:\mathbb{X}\times\mathbb{Y}\to\mathbb{R}$ is a measurable transition potential. The associated reference probability measure on $\mathbb{X}\times\mathbb{Y}$ is defined by

$$\mathcal{P}_0(d(x,y)) := (\lambda_U\times\mathcal{Q})(d(x,y)) := \lambda_U(dx)\ \mathcal{Q}(x,dy). \tag{1.3}$$

The entropic optimal transport problem, also known as the Schrödinger bridge problem, with respect to the reference measure $\mathcal{P}_0$ consists in finding

$$\mathbb{P} = \underset{\mathcal{P}\in\Pi_{U,V}}{\arg\min}\,\mathrm{Ent}(\mathcal{P}\mid\mathcal{P}_0), \quad \text{with the relative entropy} \quad \mathrm{Ent}(\mathcal{P}\mid\mathcal{P}_0) := \int \log\frac{d\mathcal{P}}{d\mathcal{P}_0}\ d\mathcal{P}, \tag{1.4}$$

where $d\mathcal{P}/d\mathcal{P}_0$ denotes the Radon-Nikodym derivative of $\mathcal{P}$ with respect to $\mathcal{P}_0$. We implicitly assume that there exists at least one $\mathcal{P}\in\Pi_{U,V}$ such that $\mathrm{Ent}(\mathcal{P}|\mathcal{P}_0)<\infty$. This condition ensures

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the existence of a solution $\mathbb{P}$ to (1.4) (cf. the seminal article by Csiszár [31], as well as Section 6 in the Lecture Notes by Nutz [85], and also the survey article by Léonard [78] and references therein). Note that the product distribution

$$(\lambda_U \otimes \nu_V)(d(x,y)) := \lambda_U(dx)\ \nu_V(dy)$$

trivially belongs to $\Pi_{U,V}$. Moreover, if $\mathcal{P} = (\lambda_U \times \mathcal{K}) \in \Pi_{U,V}$ for some Markov kernel $\mathcal{K}(x, dy)$, then

$$\frac{d\mathcal{P}}{d\mathcal{P}_0}(x,y) = \frac{d(\lambda_U \otimes \nu_V)}{d\mathcal{P}_0}(x,y) \times \frac{d\mathcal{P}}{d(\lambda_U \otimes \nu_V)}(x,y)$$

which yields the decomposition

$$\mathrm{Ent}(\mathcal{P} \mid \mathcal{P}_0) = \int W(x,y)\ \mathcal{P}(d(x,y)) + \mathrm{Ent}(\mathcal{P} \mid \lambda_U \otimes \nu_V) - \int V(y)\nu_V(dy).$$

Choosing a transition potential of the form $W(x,y) = \mathcal{W}(x,y)/t$, for some parameter $t > 0$, problem (1.4) becomes equivalent to the regularized optimal transport problem

$$\mathbb{P} = \underset{\mathcal{P} \in \Pi_{U,V}}{\arg\min} \left(\int \mathcal{W}(x,y)\ \mathcal{P}(d(x,y)) + t\ \mathrm{Ent}(\mathcal{P} \mid \lambda_U \otimes \nu_V) \right). \tag{1.5}$$

In the optimal transport literature, $\mathcal{W}$ is referred to as the cost function and $t > 0$ as the entropic regularization parameter. The limiting case $t = 0$ formally corresponds to the classical Kantorovich optimal transport problem with cost $\mathcal{W}$. For further background and developments, we refer to [49, 88, 104] and the references therein.

1.2. **Background.** The Schrödinger bridge problem (1.4) consists in constructing a random evolution that minimizes the relative entropy with respect to a given reference model while matching prescribed initial and final marginal distributions. This convex optimization problem can be interpreted as an entropic regularization of the Monge–Kantorovich optimal transport problem; we refer to [78, 85, 87] for recent surveys and comprehensive bibliographies.

Entropic optimal transport and Schrödinger bridges have become central tools in a wide range of applications, including generative modeling and machine learning [5, 36, 72, 87, 90, 91], statistical barycenters [1, 4, 10, 29, 33, 37], economics [13], computer vision [50, 99], and control theory [18, 19].

Under mild integrability conditions, the Schrödinger bridge problem admits a unique solution that can be characterized as the fixed point of the Sinkhorn equations, also known as the iterative proportional fitting procedure. We refer to the pioneering works [47, 59, 73, 97, 98] and to more recent developments in [18, 32, 75] for historical background and modern perspectives. Sinkhorn bridges involve nonlinear integral operators acting on infinite-dimensional function spaces, and explicit solutions are generally unavailable. In finite state spaces, the Sinkhorn algorithm reduces to alternating row and column normalizations of matrices. For Gaussian models, however, Sinkhorn equations can be solved analytically using sequential Bayesian conjugacy arguments [2, 29, 28, 67].

Sub-linear convergence rates of Sinkhorn equations have also been developed in the articles [3, 17, 52]. Linear rates on non-necessarily compact spaces were first obtained by Léger in [77] using elegant gradient descent and Bregman divergence techniques (see also the recent articles [36, 68]). Refined convergence rates at least one order faster for a general class of models including sub-Gaussian target marginals has also been developed in [61] under some exponential moments estimates on Schrödinger potentials. Most existing results on the exponential convergence of Sinkhorn iterations concern finite or compact state spaces, or bounded cost functions, typically relying on Hilbert's projective metric and related techniques [15, 58, 97, 98, 100, 18, 46, 60, 79]. See also [63] for refined analyses on the torus and [23] for recent advances in compact settings.

As emphasized in [25], contraction in the Hilbert metric is a very strong property, as this metric dominates all ϕ-divergences. While Hilbert metric contraction implies exponential decay for a broad class of divergences (including total variation, relative entropy, and Wasserstein distances), it is generally difficult to obtain sharp contraction estimates for specific divergence criteria. Moreover, even on totally bounded spaces such as $\mathbb{X} =]0,1[= \mathbb{Y}$, the cost function and Schrödinger potentials may be unbounded, and the associated Markov operators may have unit Birkhoff contraction coefficient; see Sections 3.2.1, 5.3, and 5.4.

Quadratic costs on Euclidean spaces provide another important class of unbounded models. In this setting, [2] develops a refined stability analysis for Gaussian Schrödinger bridges based on matrix Riccati equations. Beyond Gaussian models, only a few recent works address exponential convergence of Sinkhorn iterations for non-compact spaces and unbounded costs, notably [26, 22, 54]. The article [26] establishes quantitative exponential convergence for symmetric quadratic costs and marginal distributions with asymptotically positive log-concavity, using diffusion reflection couplings and convexity propagation techniques [27, 53]. These results were further refined in [22], which applies to all regularization parameters and certain perturbations of log-concave models. A central tool in [22] is a stability theorem controlling the entropy of Schrödinger bridges by the entropy of their marginals, under uniform semi-concavity assumptions on unknown dual Schrödinger potentials. This approach is also limited to symmetric quadratic-type costs and does not cover important classes of nonsymmetric reference transitions, such as those arising in linear Gaussian models for Ornstein–Uhlenbeck processes and denoising diffusion models. Recent work [45] partially overcomes these limitations by establishing entropic continuity results for nonsymmetric transitions and strongly convex-at-infinity potentials.

An alternative approach based on generalized Hilbert metrics for weighted function spaces is developed in [54]. Extending results in [18, 46] to unbounded cost functions, the recent article [54] presents exponential decays of these new classes of Hilbert's projective metrics with respect to some cone of functions in settings where the marginal distributions have sufficiently light tails compared to the growth of the cost function. These results also imply the exponential convergence of the total variation distance. On normed finite dimensional spaces, the main condition in [54] is that W grows faster that a polynomial of order p and the tails of both marginals decay faster that $e^{-r^{p+q}}$ for some $q > 0$. This condition doesn't cover Gaussian models.

Finally, Sinkhorn semigroups belong to the broader class of time-inhomogeneous Markov processes that share a common invariant measure at each step [2]. Stability analysis for such processes is notoriously delicate, as transition operators may vary substantially over time. These classes of time varying Markov semigroups arise in a variety of areas in applied mathematics including on nonlinear filtering, physics and molecular chemistry, see for instance [6, 38, 40, 41, 93] and references therein. While general tools exist—such as coupling methods, spectral techniques, and functional inequalities [51, 94, 95]-their direct application to Sinkhorn semigroups is challenging due to the nonlinear conjugate structure of the updates. This motivates the development of new analytical tools. In this work, we introduce a Lyapunov-based, operator-theoretic framework for the quantitative stability analysis of Sinkhorn semigroups on weighted Banach spaces.

1.3. **Contributions.** We investigate Lyapunov approaches and related weighted-norm contraction techniques to analyze the convergence of Sinkhorn semigroups. These semigroup techniques are basic quantitative methods for studying the stability of nonlinear and time-varying stochastic models, see for instance [41] as well as Section 2 in [6]. In the context of time-homogeneous models, we also refer to the pioneering articles by Hasminskii [65] and Meyn and Tweedie [80], see also the book [81] and references therein.

These powerful off-the-shelf semigroup techniques are mainly based on two conditions:

(1) The existence of a Lyapunov function satisfying a drift-type condition.